Markscheme

Mathematics: analysis and approaches
Practice question bank

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53. (a)

Consider the equation $9^x - 4 \times 3^x + 3 = 0$.

Find the sum of all values of x satisfying the equation.

let $u = 3^x$: $u^2 - 4u + 3 = 0$ *(M1)* **A1**

$(u-1)(u-3) = 0 \Rightarrow u = 1$ or $u = 3$ *(M1)*

$3^x = 1 \Rightarrow x = 0$; $3^x = 3 \Rightarrow x = 1$ **A1**

sum $= 0 + 1 = 1$ **A1**

[5 marks]